

Constrained Optimization

Design and Analysis of Algorithms

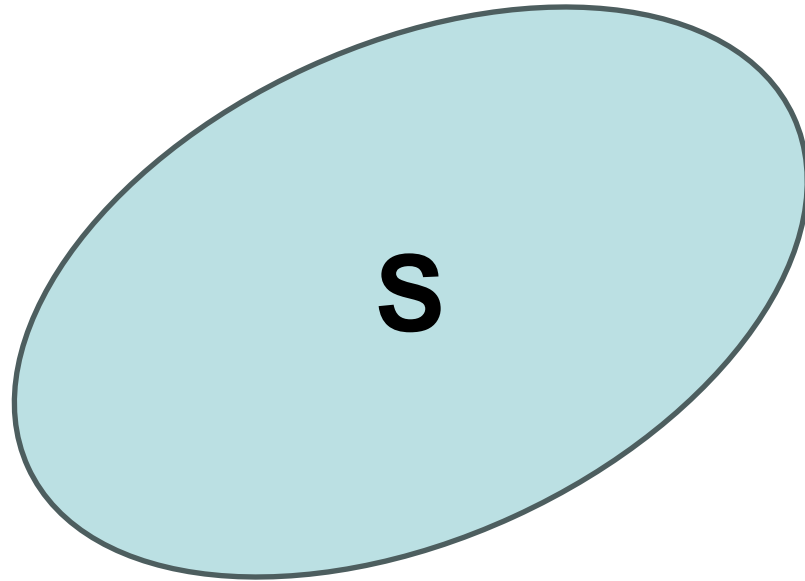
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Constrained Optimization

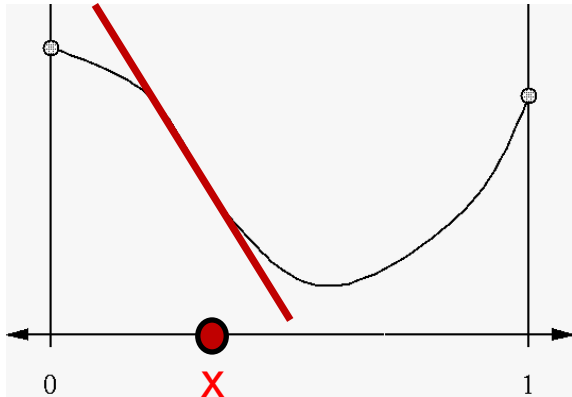
- Minimize $f(x)$ over all $x \in S$



- If f is linear and S described by linear inequality constraints, then this is a linear program.
- If f and S are both convex, problem still relatively tractable.

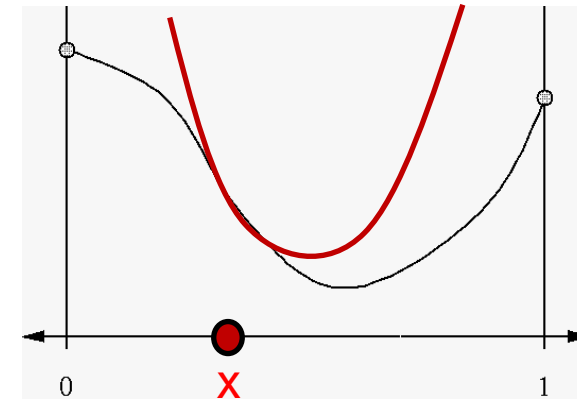
Starting Point: Unconstrained Methods

- Most involve iterative refinement, picking a direction of movement and a step size.



Gradient Descent: use first derivative to locally approximate function around x as a line / hyperplane.

$$f(x + y) \approx f(x) + y^T \nabla f(x)$$

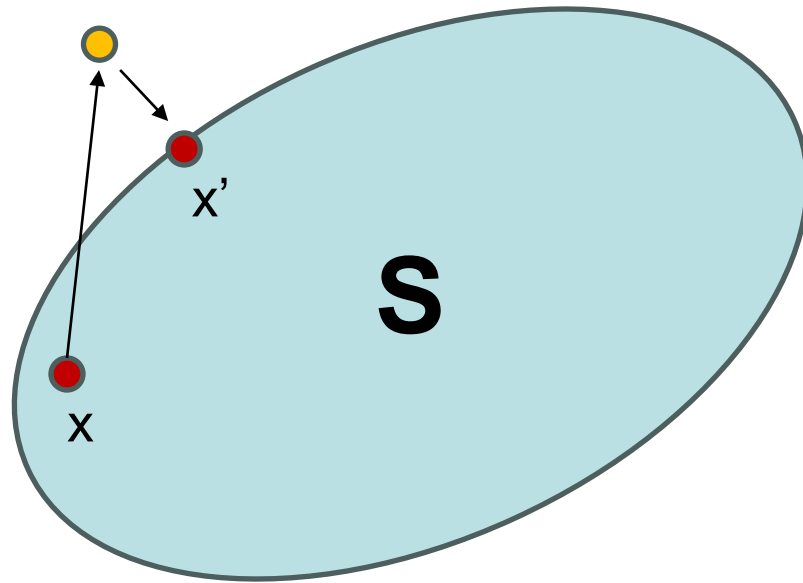


Newton's Method Use second derivative to approximate function around x locally like a quadratic.

$$f(x + y) \approx f(x) + y^T \nabla f(x) + y^T \frac{\nabla^2 f(x)}{2} y$$

Projected Gradient Descent

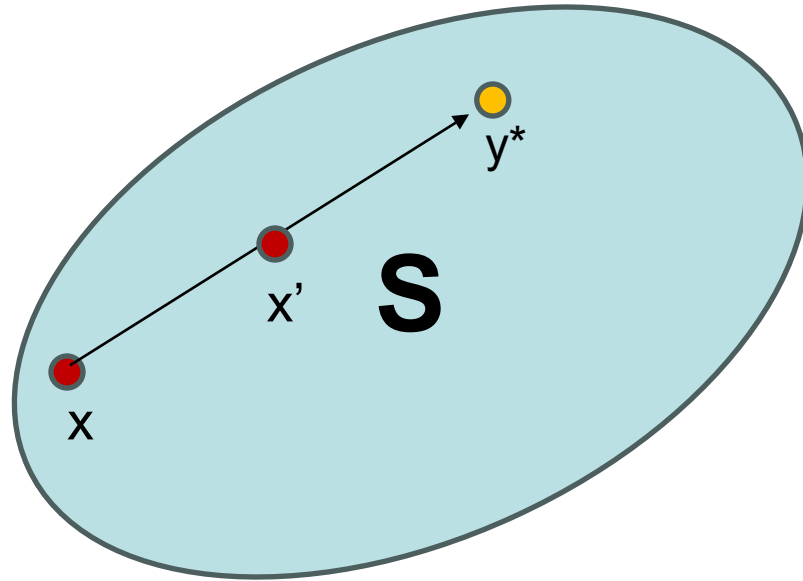
- Take gradient descent step, then project to S .



- Convergence guarantees of gradient descent preserved!
- Usually only easy to implement when S has a simple shape, enabling straightforward projection (e.g., sphere, rectangle).

The Frank-Wolfe (Conditional Gradient) Method

- Nonlinear objective $\rightarrow$ successive linear objectives



- Minimize $\nabla f(x)^\top (y - x)$ (based on linear approximation of f at x) over $y \in S$ to get y^* (a linear program, if S defined by linear constraints)
- Move in the direction of y^* with line search to find next guess x' .

Lagrangian Relaxation / Hard $\rightarrow$ Soft Constraints

- Original: Minimize $f(x)$ such that $g(x) \leq 0$.
- Relaxed: Minimize $f(x) + \lambda g(x)$, where $\lambda \geq 0$.
- Set $\lambda = 0 \rightarrow$ ignore constraint
- Set λ very high $\rightarrow$ constraint exerts too much importance
- Dual problem: how to set λ optimally
$$\max_{\lambda} d(\lambda) \leq \min_x f(x) \text{ such that } g(x) \leq 0$$
$$\text{where } d(\lambda) = \min_x f(x) + \lambda g(x)$$
- Further reading: Karush-Kuhn-Tucker (KKT) optimality conditions in this setting (generalizing $\nabla f(x) = 0$ in the unconstrained case).